



Characterization, modeling and simulation of the impact damage for polymer bonded explosives



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ABSTRACT

The impact damage behavior of polymer bonded explosives (PBXs) is critical to the safety of explosive systems. In this work, two different low velocity impact experiments were designed to study this behavior. In multi-axial loading experiments, the loading styles were deliberately designed for easier specimen recycling, avoiding secondary damage to samples. The damage mechanisms and the corresponding fracture modes of PBX under different impact loadings were analyzed. A damage constitutive model with viscoelastic response and statistical fracture for PBX was developed. The model was implemented in ABAQUS to predict the damage responses of PBX under impact loadings. Computational results were verified by the experiments.

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1. Introduction

Polymer-bonded explosives (PBXs) are usually defined as particulate composites containing energetic materials (such as HMX and RDX) embedded in a polymer binder. The content of explosive crystals in PBXs can vary from 60 to 95% by mass. Polymer binder is typically composed of plasticizers, curing catalysts, reactant, aluminum power, etc. PBXs are widely used in the aerospace engineering and defense industry. It is well known that damage in energetic materials will degrade the mechanical properties of explosives and increase the ignition sensitivity, which can be caused by factors such as impact, shock, combustion, and even the detonation of explosives [1–5]. The impact damage is crucial in the vulnerability evaluation of explosives.

Impact damage to energetic materials may arise from events such as plate impacts and Taylor tests. For impact ignition mechanisms of energetic material, the results from Ho et al. [6] based on the modified Hopkinson bar showed that the impact sensitivity and ignition mechanisms are mostly determined by the dynamic mechanical properties of energetic materials. On the impact damage mechanisms of PBX, studies in [7,8] of high-speed projectiles and flying plates showed that impact loading induces many micro-cracks and micro-voids; however, they may also induce impact damage under both one-dimensional planar and two-dimensional axisymmetric loadings. To overcome this, we developed a multi-axial impact loading experiment and used it to apply dynamic loading.

Many efforts have been made to address the damage modes of energetic materials by means of experiments. For example, Seaman

et al. [9,10] designed impact experiments and studied the fracture mechanisms of energetic materials under high strain rates. Their work showed that a micro fracture model containing nucleation, growth, and coalescence processes for cracks is effective in describing dynamic fracture, spall, and fragmentation in energetic materials. Asay et al. [11,12] designed a non-shock ignition experiment in which a small piece of confined PBX9501 was impacted by a steel plunger and the displacements and temperatures on the surface were measured. Chen et al. [13,14] studied the damage modes and fracture mechanisms under diametric compression, creep, quasi static tension and compression tests. Their results showed the damage modes and fracture mechanisms were affected by the loading conditions. Rae et al. [15,16] studied the deformation and failure of PBX9501 using the Brazilian test and optical microscopy. They found that the failure paths tend to run around the long straight edges of the explosive filler and avoid regions of fine filler and binder. The evolution of the microstructural damage can be considered as the deboning of large particles from the matrix, grain splitting, binder crack, void nucleation and growth. As shown in Fig. 1, severe stressing and extreme temperatures induce the damage modes of energetic materials. It has been shown that localized sudden damage can cause the formation of hot spots and thus trigger the detonation of high explosives under low-level loading.

Dienes [17] showed that the main contribution to potential heat formation is the frictional force on shear crack surfaces. Howe et al. [18] studied the role of shear in the initiation of detonation through impact experiments. They showed that shear cracking certainly occurs in energetic materials under impact loading. Dienes [19,20] originally developed the statistical crack mechanical model (SCRAM) to model the dynamic deformation and fragmentation of brittle materials. Damage in his model was described by the formation of

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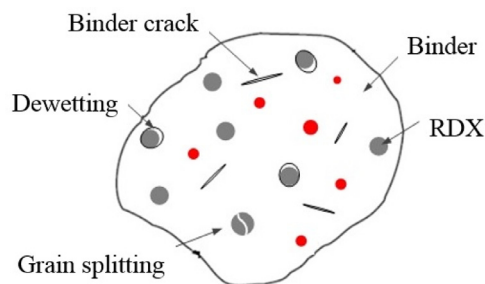


Fig. 1. Schematic view of a PBX microstructure with various damage modes.

penny-shaped cracks and the analytical solution for the deformation fields of these cracks was used to describe the damage component in the total deformation rate tensor. Based on SCRAM, a visco statistical crack mechanical model (Visco-SCRAM) was developed to investigate the mechanical behavior and ignition of PBX9501 under non-shock impacts [21,22]. The predictions in static and dynamic compression agreed well with the experimental results. The model was also used to predict hot spot generation in PBX 9501 under dynamic impacts. The simulated results are consistent with those from the non-shock ignition experiment by Asay et al. [11,12].

In the current work, two types of gas-gun impact experiments were performed to study the damage modes and failure mechanisms of PBX. One is the uniaxial impact test for investigating the macro fracture modes under two-dimensional axisymmetric loadings. The other is the multi-axial impact test for study of micro damage modes and the fracture mechanisms. The methodology of the experimental verification and comparison of experimental results with computational predictions in PBX under impact loading are given below.

2. Experimental analysis

2.1. PBX composition

PBX is a heterogeneous material consisting of 60 wt. % energetic hexahydro-1, 3, 5-trinitro-1, 3, 5-s-triazine (RDX) embedded in 40 wt. % polymer binder. The polymer binder consists of 24 wt. % HTPB (hydroxy-terminated polybutadiene) and 16 wt. % aluminum powder. Nanometer-size aluminum powder is known to react more rapidly than conventional additives. The sizes of RDX are relatively uniform at approximately $100\ \mu\text{m}$. The mass density of PBX is $1.69\ \text{g/cm}^3$.

2.2. Uniaxial impact tests

To study the macroscopic failure mechanisms of PBX without radial constraints, a cylinder of PBX connected to a projectile was propelled to a range of speeds from 20 m/s to 130 m/s. As shown in Fig. 2(a), the experimental facility mainly consists of a light gas gun, projectiles, impact anvil and a velocity measurement chamber. A hollow projectile is used to reduce the weight of the projectile. The projectile is made of aluminum and weighs 0.027 kg, as shown in Fig. 2(b). The impact stresses of the samples were measured by polyvinylidene fluorid (PVDF) sensors. The size of the PVDF sensors are $8\ \text{mm} \times 8\ \text{mm} \times 0.1\ \text{mm}$. The PBX samples with a diameter of 20 mm and a thickness of 5 mm were fabricated via a pouring process. One pair of laser light curtains are used to trigger the timing device. The interval measured is used to calculate the velocities of the projectiles.

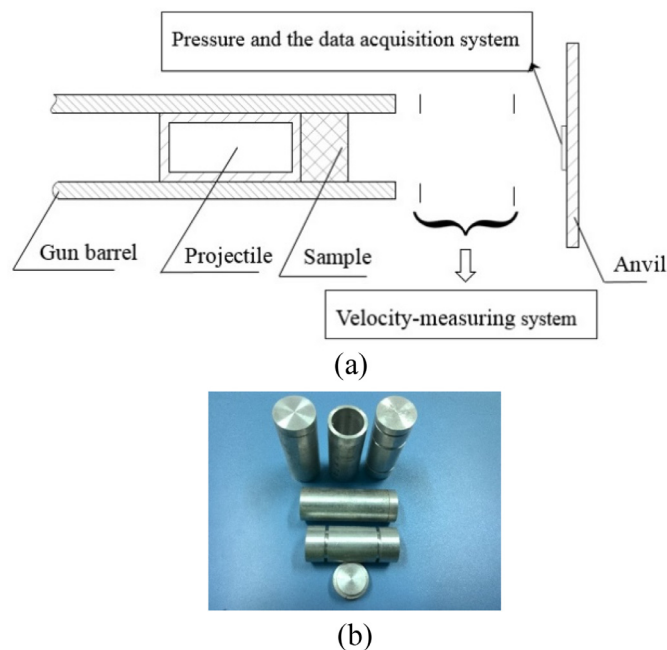


Fig. 2. (a) Schematic view of the uniaxial impact experiment facility, (b) hollow projectiles (16 mm ID, 20 mm OD, 60 mm long).

2.3. Multi-axial impact tests

Fig. 3 shows the multi-axial impact experimental system. The test system mainly consists of a light gas gun, projectiles, loading equipment, fixed device, target warehouse and a velocity-measuring chamber.

The loading device is shown in Fig. 4(a). The PBX sample is 20 mm in diameter and 5 mm in thickness, as shown in Fig. 4(b). The lengths of projectiles are 60, 100 and 120 mm with the same diameter of 20 mm as the PBX sample. All the projectiles were made of hardened Q235 steel. Figs. 5(a) and (b) show the disassembled and assembled equipment, respectively. Figs. 5(c) and (d) show the steel tube and the wedge-shaped semicircular grooves, respectively. Steel bar-1 has a diameter of 20 mm and a length of 150 mm, while bar-2 has a diameter and length of 50 mm and 110 mm, respectively. Encapsulated in two wedge-shaped semicircular grooves with a wrapping steel ring, the specimen is loaded by steel bar-1, as shown in Fig. 5(b). The steel tube was locked by a nut. The strain gauges were glued onto the external surface of bar-2 to record the stress history during the impact process.

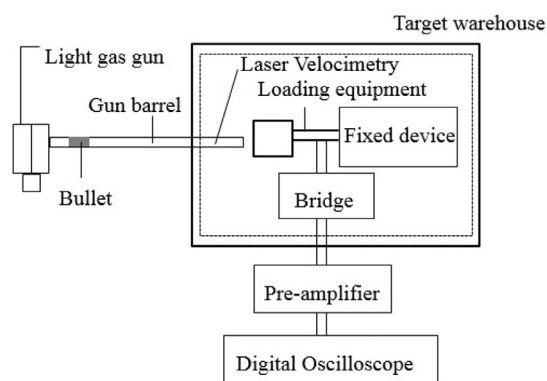


Fig. 3. Schematic diagram of test system.

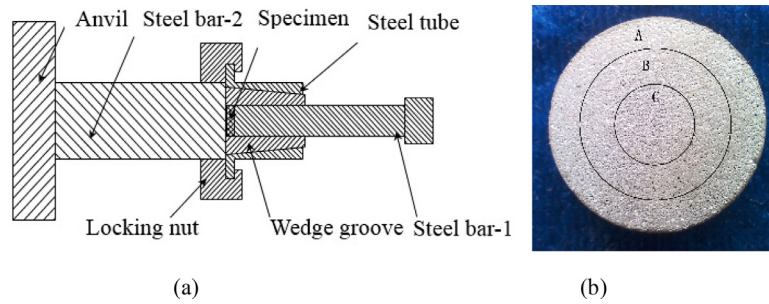


Fig. 4. (a) Experimental facility of dynamic loading and (b) sample.

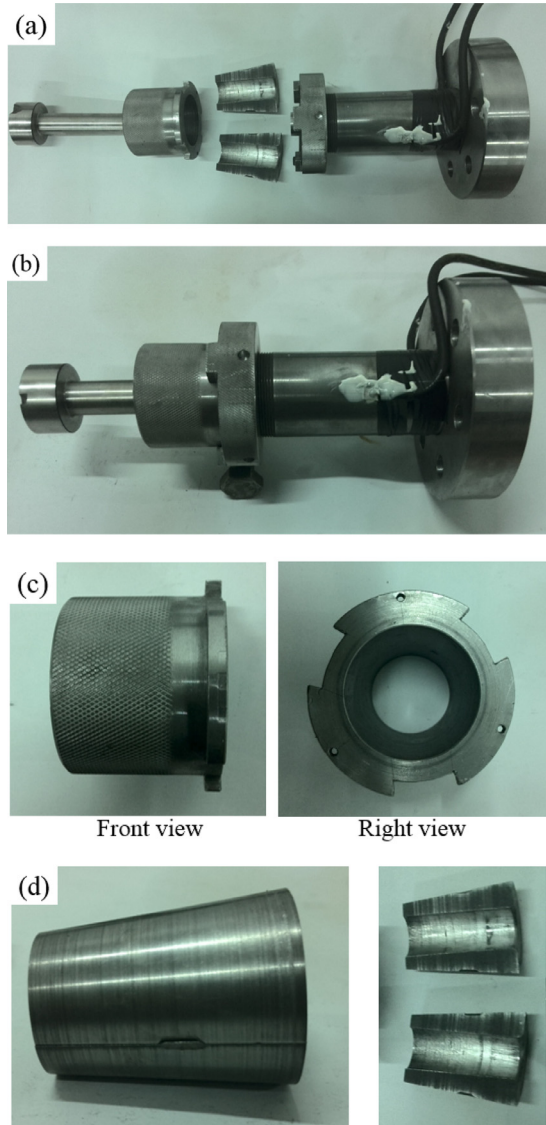


Fig. 5. (a) The disassembled equipment, (b) the assembled equipment, (c) steel tube, and (d) wedge-shaped semicircular grooves.

3. Experimental analysis and discussion

3.1. Material damage without confinement

The impact conditions and corresponding damages in the targets are summarized in Table 1.

Fig. 6 demonstrates the surfaces of the samples from the uniaxial impact tests. The damage became more and more severe with

Table 1

Experimental results of the samples without confinement.

Samples	Impact velocity (m/s)	Peak stress (MPa)	Observed fracture results
1	20.5	17.8	No damage
2	42.8	32.4	Formation of the initial crack
3	—	37.5	Obvious crack coalescence
4	63.5	45.9	Partially spalled
5	90.5	71.5	Completely fragmented
6	110.5	88.5	Completely fragmented

increasing impact energy. At 20.5 m/s, as shown in Fig. 6(a), no visible damage was observed in the sample. At 42.8 m/s (Fig. 6(b)), obvious cracks were observed in the edge area of the specimen. At 90.5 m/s (Fig. 6(c)), the specimen was completely fragmented, while at 110.5 m/s, the sample was completely damaged and could not be recycled. In these tests, the crack plane is normal to the direction of propagation, and there is little shearing response. These cracks are caused by tensile stress arising from the rarefaction waves propagating from the free cylindrical surfaces toward the cylinder axis.

It was observed that the tensile strength of the PBX is less than the compressive strength. This is due to the difference in the micro damage mechanisms. It appears that even under extremely low stress, the initial voids and crack may be initiated in the PBX. Under the tensile loading, these voids and crack coalesce so that the intensity becomes much lower than that in compression.

3.2. Material damaged under multi-axial loading

The mass densities variation (before and after the impact loading), impact velocities, and peak stresses are summarized in Table 2. The density of the samples decreases due to the presence of extensive micro-cracks. Both density variation and damage to the PBX samples increase with higher impact velocities. Figs. 7 and 8 depict the density variation before and after the impact and the recorded stress history curves of the PBX specimens under different impact loading conditions. The highest impact stress recorded was 128.1 MPa, leading to a density decrease of 10.9%. Under similar peak stress, longer bullets turn to give more density reduction. The duration of stress waves with different projectile lengths are shown in Fig. 8. The pulse intervals are wider with longer projectiles.

The results for Samples 2 and 15 under similar impact velocities are given in Table 2. The peak stresses of Samples 2 and 15 are 29.075 MPa and 102.12 MPa, respectively. The results suggest that increasing the bullet length can lead to a higher peak stress and more serious damage under similar impact velocities. In the cases of the similar peak stresses but different bullet lengths, more serious damage can be observed in the samples with a longer bullet.

The fractography was observed by SEM. The micro-cracks are shown in Fig. 9. Fig. 9(a) shows the undamaged specimens, while

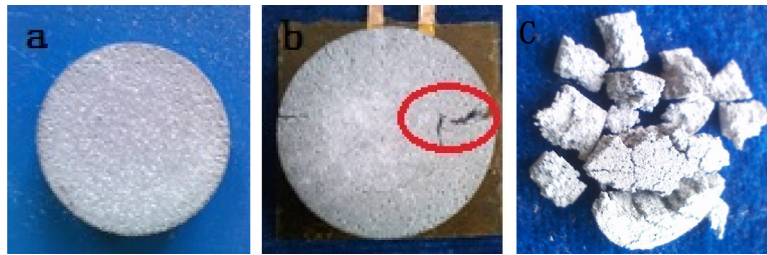


Fig. 6. Samples after uniaxial impact tests.

Table 2
Results of multi-axial impact tests.

Sample	L_s (mm)	Impact velocity (m/s)	Peak stress (MPa)	Density change (%)
1	60	17.1	12.331	−0.05
2	60	31.8	29.075	−0.57
3	60	42.1	60.358	−4.12
4	60	55.8	87.255	−6.15
5	60	77.8	128.13	−10.9
6	100	20.5	22.234	−0.99
7	100	25.2	32.993	−1.8
8	100	29.6	52.965	−3.3
9	100	35.8	66.758	−4.8
10	100	46.2	79.834	−6.8
11	100	53.8	98.4	−9.14
12	120	16.8	26.841	−2.2
13	120	21.4	49.6	−5.30
14	120	25.6	69.379	−7.8
15	120	32.4	102.12	−10.32

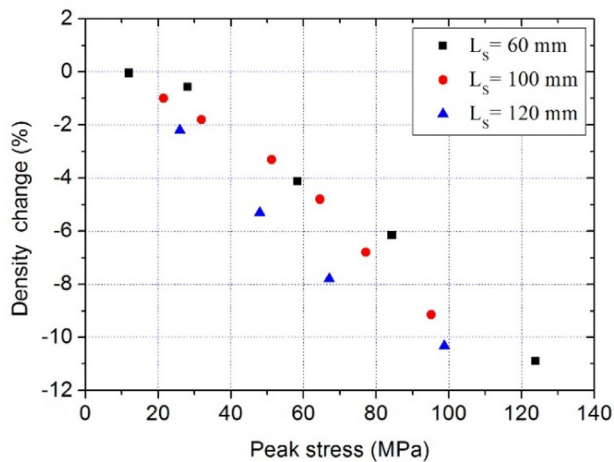


Fig. 7. Density change of the PBX specimens before and after impact under different impact loading conditions.

Figs. 9(b), (c) and (d) show damaged ones with crack lengths of approximately $99.9 \mu\text{m}$, $204 \mu\text{m}$ and $269 \mu\text{m}$, respectively. The results indicate that the increment of the impact velocity can cause crack growth.

3.3. Fractography study

The fractured surfaces in uniaxial tests were observed by SEM, as shown in Fig. 10. The intact large crystals were clearly pulled out from the matrix, indicating the main damage mode is the trans-granular failure in the uniaxial impact tests. These initial failure sites may extend to form large, irregular and planar crack-like features.

At multi-axial impact loading, interface debonding between the grain and polymer binder is the predominant failure mode, as shown

in Fig. 11(a). Crystal cleavage is also observed, as shown in Figs. 11(b) and 12. Before merging into larger cracks, the failure generally initiates at many separate sites, usually around the boundaries of large filler particles. Due to the typical inhomogeneous microstructure of PBX, interfacial shear stress can be generated even under compression. It is assumed here that the cavitation within the rubber causes debonding under multi-axial impact loading. The polymer binder is a rubbery material. It obeys the viscoelasticity constitutive law and serves as a cushion for the grain. The energetic grains are brittle at ambient pressures and therefore cannot undergo plastic deformations. This leads to particle cracking accompanying the particle-matrix interface debonding. The individual cavitation can coalesce to form a failure front.

Due to the constraints of the fixture, both expansion and pore opening are limited, leading to continuous increases in the internal stresses and even fracturing of local regions. Fig. 13 shows the SEM images of the damaged PBX at different locations. Figs. 13(a), (b) and (c) correspond to the regions marked with A, B and C in Fig. 4(b), respectively. The number densities of the micro-cracks at the three typical regions are different, being higher in region C. It should be stressed that the impact induced damage is not homogenous in the sample. The central area contains more damage.

Fig. 14 shows the SEM images of the cracks. Fig. 14(a) displays interconnected cracks at the surface of the PBX sample. Fig. 14(b) zooms into the finer structure of selected cracks, revealing a complex nature of porosity coalescence. Fig. 14(c) provides further evidence of the tubular (or one dimensional) nature of the porosity. Due to the limitations of the SEM technique, the depth of the crack cannot be determined.

4. Visco elastic constitutive model (Viaco-SCRAM)

4.1. Model formulation

The model formulation presented here is based on the work of Addessio and Johnson [23], incorporating a generalized Maxwell

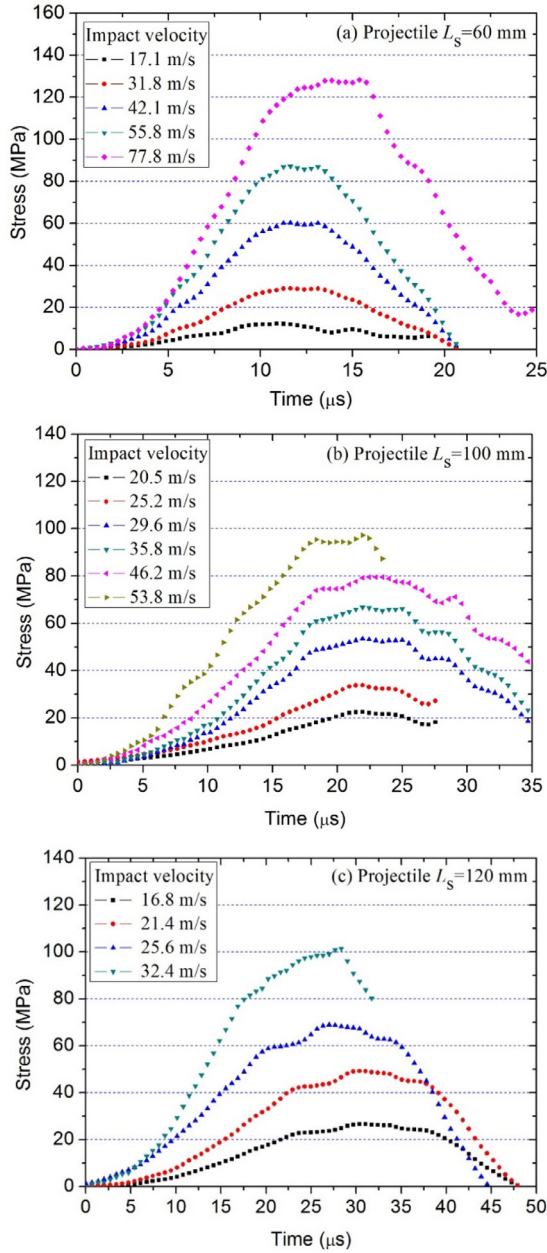


Fig. 8. Stress history curves under different load conditions.

model (GMM) with the SCRAM approach of Dienes [19,20]. An isotropic constitutive relation is employed to describe the damage response of PBX. During the process of deformation, it is assumed that the crack distribution remains random and the size distribution of the cracks is exponential.

The stress can be decomposed into deviatoric and mean components as

$$\dot{\sigma}_{ij} = \dot{S}_{ij} + \dot{\sigma}_m \delta_{ij} \quad (1)$$

In a Maxwell element, the relationship between deviatoric stress and strain in the elastic element is

$$S_{ij}^e = 2G e_{ij}^e \quad (2)$$

where G is the shear modulus, and the relationship between the stress and the viscous strain rate is

$$S_{ij}^{ve} = 2\eta \dot{e}_{ij}^v \quad (3)$$

where η is the viscosity of the dashpot, and a dot over a variable indicates the derivative of the variable with respect to time. In a Maxwell model, the kinematic condition can be determined as

$$e_{ij}^{ve} = e_{ij}^e + e_{ij}^v \quad (4)$$

where e_{ij}^{ve} is the total deviatoric strain in the Maxwell element, e_{ij}^e is the deviatoric strain in the spring, and e_{ij}^v is the deviatoric strain in the damper. Combining Eqs. (2), (3) and (4), the strain rate can be determined as

$$\dot{e}_{ij}^{ve} = \frac{\dot{S}_{ij}}{2G} + \frac{S_{ij}}{2\eta} \quad (5)$$

The deviatoric stress rate in each Maxwell component is given by Eq. (6)

$$\dot{S}_{ij}^{(n)} = 2G^{(n)} \dot{e}_{ij}^{ve} - \frac{S_{ij}^{(n)}}{\tau^{(n)}} \quad (6)$$

where $\tau = \eta/G$.

In the Visco-SCRAM model, the deviatoric strain rate is the sum of the deviatoric viscoelastic strain rate and the deviatoric cracking strain rate

$$\dot{e}_{ij} = \dot{e}_{ij}^{ve} + \dot{e}_{ij}^c \quad (7)$$

Combining Eqs. (6) and (7) gives

$$\dot{S}_{ij}^{(n)} = 2G^{(n)} (\dot{e}_{ij} - \dot{e}_{ij}^c) - \frac{S_{ij}^{(n)}}{\tau^{(n)}} \quad (8)$$

The equilibrium constraint provides that the solution for the global stress is a sum of the individual stresses,

$$\dot{S}_{ij} = \sum_{n=1}^N \dot{S}_{ij}^{(n)} \quad (9)$$

The general relationship between the deviatoric stress, stress rate and strain rate is assumed to be

$$\dot{S}_{ij} = \sum_{n=1}^N \dot{S}_{ij}^{(n)} = \sum_{n=1}^N \left(2G^{(n)} \dot{e}_{ij}^{ve} - \frac{S_{ij}^{(n)}}{\tau^{(n)}} \right) \quad (10)$$

The deviatoric strain rate is

$$\dot{e}_{ij}^{ve} = \frac{\dot{S}_{ij} + \sum_{n=1}^N \frac{S_{ij}^{(n)}}{\tau^{(n)}}}{\sum_{n=1}^N 2G^{(n)}} = \frac{\dot{S}_{ij} + \sum_{n=1}^N \frac{S_{ij}^{(n)}}{\tau^{(n)}}}{2G_{total}} \quad (11)$$

where $G_{total} = \sum_{n=1}^N G^{(n)}$. The fundamental deviatoric cracking strain versus deviatoric stress relationship is shown in Addessio and Johnson [23] as

$$e_{ij}^c = \beta c^3 S_{ij} \quad (12)$$

where c is the average crack radius, and β is a parameter relating the shear modulus and the initial crack distribution N_0 as

$$2G\beta = AN_0 = 1/a^3 \quad (13)$$

where A and a are constants and the initial flaw size, respectively. With this definition, the relationship between the deviatoric strains and the crack radius is

$$2G_{total} e_{ij}^c = \left(\frac{c}{a} \right)^3 S_{ij} \quad (14)$$

The deviatoric cracking strain rate is

$$\dot{e}_{ij}^c = \frac{1}{2G_{total}} \left[\left(\frac{c}{a} \right)^3 \dot{S}_{ij} + 3 \left(\frac{c}{a} \right)^2 \frac{\dot{c}}{a} S_{ij} \right] \quad (15)$$

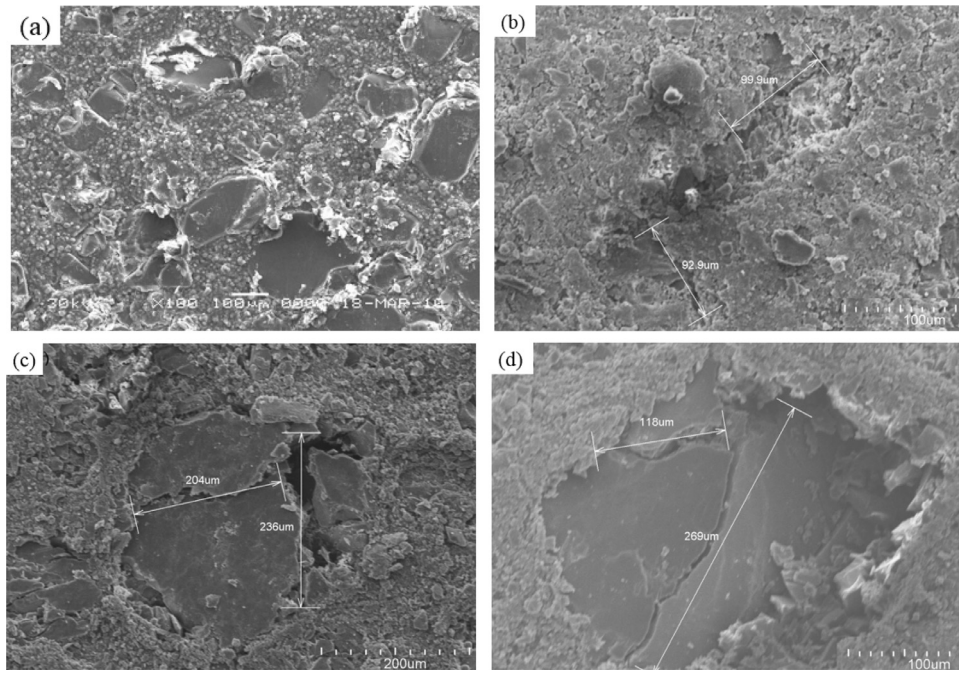


Fig. 9. The micro crack lengths in different cases: (a) no damage; (b) $L_s=60$ mm, $v=31.8$ m/s; (c) $L_s=60$ mm, $v=42.1$ m/s; (d) $L_s=60$ mm, $v=55.8$ m/s.

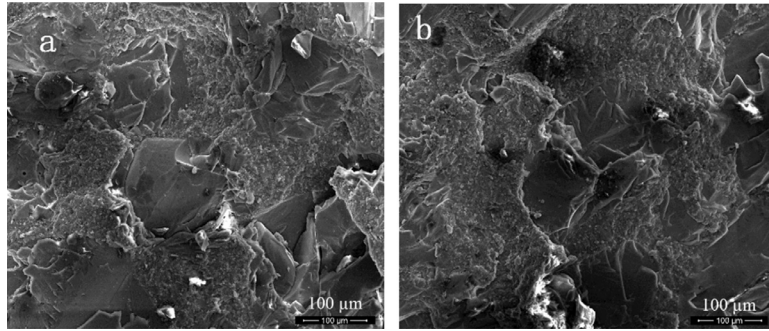


Fig. 10. A pair of fracture modes showing trans-granular failure under uniaxial impact.

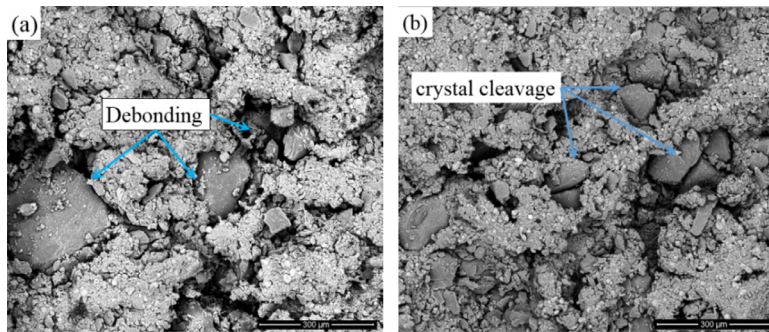


Fig. 11. SEM images of the damage surfaces of the PBX samples in the multi-axial impact tests ($L_s=100$ mm and $v=46.2$ m/s).

The deviatoric stress rate in the Visco-SCRAM model is

$$\dot{S}_{ij} = \frac{2G_{total}\dot{\epsilon}_{ij} - \sum_{n=1}^N \frac{S_{ij}^{(n)}}{\tau^{(n)}} - 3\frac{1}{a}\left(\frac{\xi}{a}\right)^2 \dot{C}S_{ij}}{1 + \left(\frac{\xi}{a}\right)^3} \quad (16)$$

The stress rate in each Maxwell element is

$$\dot{S}_{ij}^{(n)} = 2G^{(n)}\left(\dot{\epsilon}_{ij} - \frac{1}{2G_{total}}\left[\left(\frac{C}{a}\right)^3 \dot{S}_{ij} + 3\frac{1}{a}\left(\frac{C}{a}\right)^3 \dot{C}S_{ij}\right]\right) - \frac{S_{ij}^{(n)}}{\tau^{(n)}} \quad (17)$$

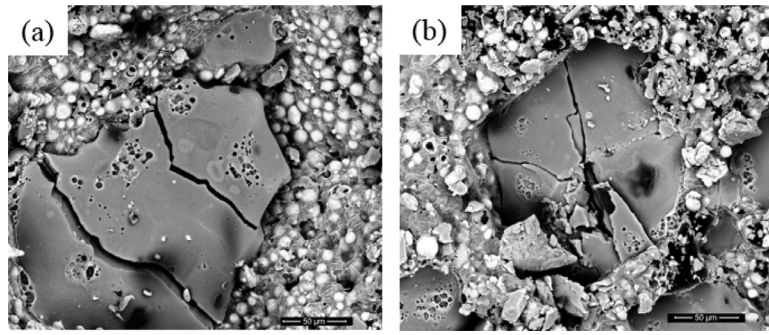


Fig. 12. SEM images of the crystal cleavage of the PBX samples in the multi-axial impact tests ($L_s=60$ mm and $v=42.1$ m/s).

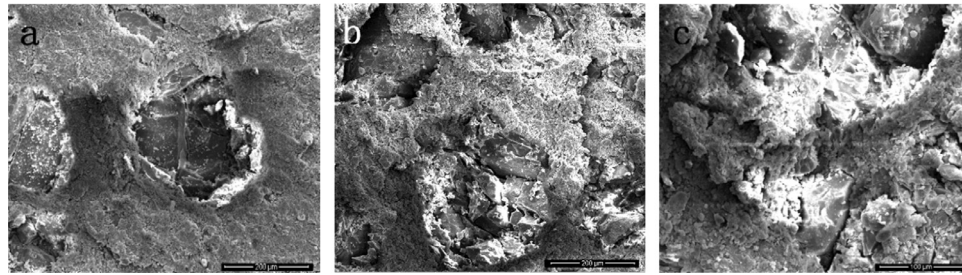


Fig. 13. An SEM image of the fracture surfaces of the PBX in the multi-axial impact tests ($L_s=100$ mm and $v=46.2$ m/s).

Based on the work of Dienes [24], we assumed that the crack growth rate depends on the stress intensity. The crack radius growth rate is determined as a function of K , namely,

$$\dot{c} = v_R \left(\frac{K}{K_1} \right)^m \quad K < K_0 \sqrt{1 + \frac{2}{m}} \quad (18)$$

$$\dot{c} = v_R \left[1 - \left(\frac{K_0}{K} \right)^2 \right] \quad K \geq K_0 \sqrt{1 + \frac{2}{m}} \quad (19)$$

where $K_1 = K_0 \sqrt{1 + \frac{2}{m} \left(1 + \frac{2}{m} \right)^{\frac{1}{m}}}$. The relation between K and the Von Mises equivalent stress is

$$K^2 = \frac{3}{2} \pi c S_{ij} S_{ij} \quad (20)$$

4.2. Implementation of the Visco-SCRAM in ABAQUS

The Visco-SCRAM was implemented in the commercial software package ABAQUS via the user material subroutine (VUMAT). If the

coefficients are relatively constant over a time step, they may be integrated in a closed form over the time step. Alternatively, a fourth-order Runge–Kutta scheme or Euler integration can be used. All these methods are tested and essentially show no difference for the time step size required to model the transient mechanical response of the PBX.

Combining Eqs. (16) and (17) gives

$$\dot{S}_{ij}^{(n)} = (2G^{(n)} - \lambda) \dot{e}_{ij} - \frac{S_{ij}^{(n)}}{\tau^{(n)}} + \frac{\zeta}{2G_{total}} \sum_{n=1}^N \frac{S_{ij}^{(n)}}{\tau^{(n)}} + \left(\frac{\zeta}{2G_{total}} + 1 \right) 3 \frac{1}{a} \left(\frac{c}{a} \right)^2 \dot{c} S_{ij} \quad (21)$$

where $\zeta = \frac{c^3}{a^3 + c^3}$, $\lambda = \zeta \left(\frac{c}{a} \right)^3$, $S_{ij} = \sum_{n=1}^N S_{ij}^{(n)}$.

Eq. (21) contains n differential equations. The average crack radius c is calculated by Eqs. (18), (19) and (20). The n equations can be solved by the single step fourth order Runge–Kutta method. It is important to note that the average crack radius c at time $t + \Delta t$ can be calculated by the stress S_{ij} at time t . With this algorithm, the stress components at time $t + \Delta t$ can be obtained by solving Eq. (21).

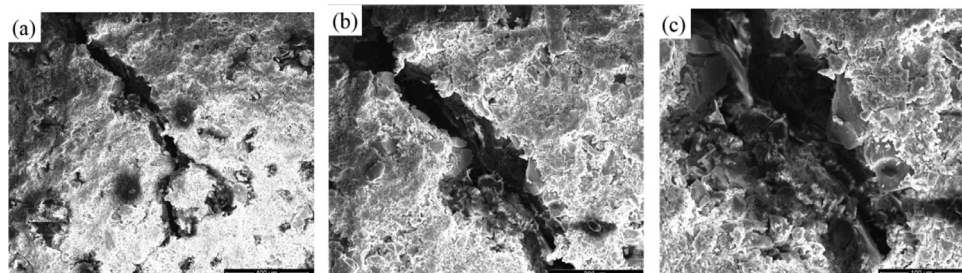


Fig. 14. SEM images of the PBX impact damage ($L_s=120$ mm and $v=32.4$ m/s).

The fourth order Runge–Kutta numerical method can be described by

$$\dot{S}_{ij}^{(n)}(t + \Delta t) = \dot{S}_{ij}^{(n)}(t) + \frac{1}{6} (K_1^{(n)} + K_2^{(n)} + K_3^{(n)} + K_4^{(n)}) \quad (22)$$

where,

$$\begin{aligned} K_1^{(n)} &= (2G^{(n)} - \lambda) \dot{\epsilon}_{ij} - \frac{S_{ij}^{(n)}}{\tau^{(n)}} + \frac{\zeta}{2G_{total}} \sum_{n=1}^N \frac{S_{ij}^{(n)}}{\tau^{(n)}} \\ &\quad + \left(\frac{\zeta}{2G} + 1 \right) 3 \frac{1}{a} \left(\frac{c}{a} \right)^2 \dot{c} S_{ij} \\ K_2^{(n)} &= (2G^{(n)} - \lambda) \dot{\epsilon}_{ij} - \frac{S_{ij}^{(n)} + \frac{1}{2} \Delta t K_1^{(n)}}{\tau^{(n)}} + \frac{\zeta}{2G_{total}} \sum_{n=1}^N \frac{S_{ij}^{(n)} + \frac{1}{2} \Delta t K_1^{(n)}}{\tau^{(n)}} \\ &\quad + \left(\frac{\zeta}{2G} + 1 \right) 3 \frac{1}{a} \left(\frac{c}{a} \right)^2 \dot{c} \sum_{n=1}^N \left(S_{ij}^{(n)} + \frac{1}{2} \Delta t K_1^{(n)} \right) \\ K_3^{(n)} &= (2G^{(n)} - \lambda) \dot{\epsilon}_{ij} - \frac{S_{ij}^{(n)} + \frac{1}{2} \Delta t K_2^{(n)}}{\tau^{(n)}} + \frac{\zeta}{2G_{total}} \sum_{n=1}^N \frac{S_{ij}^{(n)} + \frac{1}{2} \Delta t K_2^{(n)}}{\tau^{(n)}} \\ &\quad + \left(\frac{\zeta}{2G} + 1 \right) 3 \frac{1}{a} \left(\frac{c}{a} \right)^2 \dot{c} \sum_{n=1}^N \left(S_{ij}^{(n)} + \frac{1}{2} \Delta t K_2^{(n)} \right) \\ K_4^{(n)} &= (2G^{(n)} - \lambda) \dot{\epsilon}_{ij} - \frac{S_{ij}^{(n)} + \frac{1}{2} \Delta t K_3^{(n)}}{\tau^{(n)}} + \frac{\zeta}{2G_{total}} \sum_{n=1}^N \frac{S_{ij}^{(n)} + \frac{1}{2} \Delta t K_3^{(n)}}{\tau^{(n)}} \\ &\quad + \left(\frac{\zeta}{2G} + 1 \right) 3 \frac{1}{a} \left(\frac{c}{a} \right)^2 \dot{c} \sum_{n=1}^N \left(S_{ij}^{(n)} + \frac{1}{2} \Delta t K_3^{(n)} \right) \end{aligned}$$

The increment of the deviatoric stress is

$$\Delta S_{ij}^{(n)} = S_{ij}^{(n)}(t + \Delta t) - S_{ij}^{(n)}(t) = \frac{1}{6} (K_1^{(n)} + K_2^{(n)} + K_3^{(n)} + K_4^{(n)}) \quad (23)$$

$$\Delta S_{ij} = \sum_{n=1}^N \Delta S_{ij}^{(n)} \quad (24)$$

$$S_{ij}^{new} = S_{ij}^{old} + \Delta S_{ij} \quad (25)$$

where S_{ij}^{new} is the deviatoric stress at the current step, and S_{ij}^{old} is the deviatoric stress at the previous step.

4.3. Application of the model to experiments

The viscoelastic response of the PBX is modeled as a generalized Maxwell model having an elastic spring in parallel with four Maxwell elements. The master relaxation modulus curve of PBX was obtained from the references [25]. The key is to find G_i (relaxation modulus coefficients) and τ_i (relaxation time) to fit the experimental data. These parameters can be obtained by the least square algorithm. Note that G_i are always positive, so the least square algorithm is accomplished via an iterative Levenberg-Marquadt method. The shear moduli of the Maxwell elements are given in Table 3.

The assumed initial values of the internal flaw size, average crack radius, K_0 and the cracking parameter are 0.001 m, 0.00002 m, 5.0 e5 Pa, and 10, respectively. Cracking parameters are given in Table 4. The maximum value of the growth rate of the average crack radius is assumed to be 300.0 m/s.

The first example application of the material model is a characterization of the mechanical properties of the PBX. In our previous

Table 3
The relaxation modulus for the Maxwell element of the PBX.

	1	2	3	4	5
G (MPa)	472	115.94	83.56	4.27	35
τ (s)	7.5E-7	7.5E-6	7.5E-5	7.5E-4	0

Table 4
Cracking parameters.

ν	m	a (m)	c (m)	v (m/s)	K (Pa · m ^{1/2})
0.49	10	0.002	0.00002	300	2.5×10^5

work, the dynamic compressive mechanical properties of the PBX were obtained by the modified split Hopkinson pressure bar (SHPB) technique [25]. The specimens did not produce visible cracks at the strain rate of 891 s⁻¹ and 1298 s⁻¹ in the SHPB tests. In the PBX specimens tested at 2000 s⁻¹ and higher strain rates, however, many visible cracks appear perpendicular to loading direction. All these uniaxial loading tests are simulated by the Visco-SCRAM model using ABAQUS. The model configuration is shown in Fig. 15. The specimen is 4 mm long and 16 mm in diameter. Pulse-shaping techniques, which employ a pulse shaper at the end of the incident bar, have recently been developed to generate an initially slowly ascend-

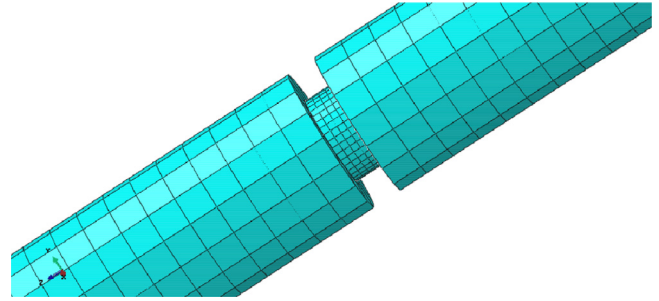


Fig. 15. Finite element meshes.

ing loading profile in the incident pulse so that the specimen can obtain the dynamic stress equilibrium in the elastic deformation stage. The loading pulse was applied at the free end of the input bar as surface pressure history. The incident wave is shown in Fig. 16. It was found that above 120 μ s, the stress decreases in the SHPB tests. The reason is that the time of incident wave is too limited, and the specimen strain cannot reach a maximum value at the strain rate of 891 s⁻¹ and 2000 s⁻¹.

The simulated and experimental stress-strain curves are shown in Fig. 17. The experimental data are taken from Ref. [25]. The good agreement validates the constitutive model proposed in this work for PBX.

The second example is the simulation of an SHPB test at the strain rate of 2330 s⁻¹, 2700 s⁻¹ and 3200 s⁻¹. The model configuration and incident wave are shown in Figs. 15 and 16, respectively. The results

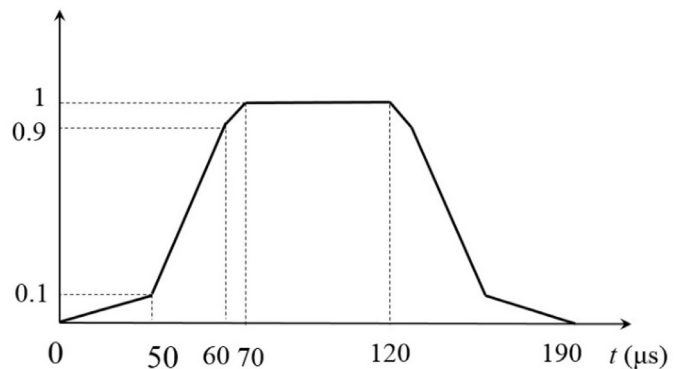


Fig. 16. Shaped input stress pulse. The incident wave is given in Ref. [25].

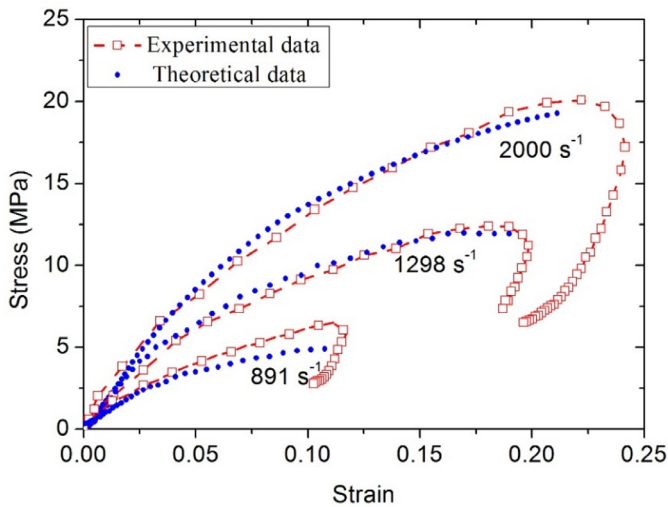


Fig. 17. Comparison of experimental [25] and predicted stress-strain.

of the axial stress-strain curves from numerical simulations and experiments at the strain rate of 2330 s^{-1} are shown in Fig. 18. The tests were conducted to determine the damage properties of the material. The predicted result at the strain rate of 2330 s^{-1} was in good agreement with the experiment. The simulated stress-strain curve shows that a stress increase range with strain is followed by a decrease range. From the simulated results at 2330 s^{-1} , 2700 s^{-1} and 3200 s^{-1} , it is also found that the strength of PBX increases gradually with the strain rate. The predicted strengths of PBX were 27.5 MPa and 32.3 MPa at 2700 s^{-1} and 3200 s^{-1} , respectively. The simulation results are in good agreement with experimental observations. When the micro crack extends to a certain length, the specimen loses bearing capacity.

The third example is the simulation of multi-axial impact tests. This aims to investigate the damage behavior of the PBX under multi-axial impact loading. The dynamic impact experiments were conducted and reported in Section 2.3. A three-dimensional finite element model (see Fig. 19) of the multi-axial test was constructed using the solid element SOLID164, with Z-axis being the longitudinal axis of the bar. Fig. 20(a) shows the comparison of the predicted and the experimental axial dynamic compressive stress-time curves at the nominal velocity of 31.4 m/s. The two curves are well matched. The crack radius is provided in Fig. 20(b). The mean crack radius

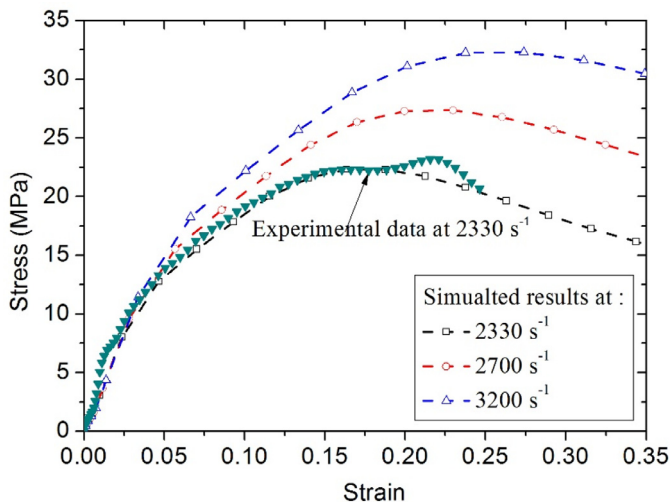


Fig. 18. Visco-SCRAM responses of the PBX with uniaxial stress-strain.

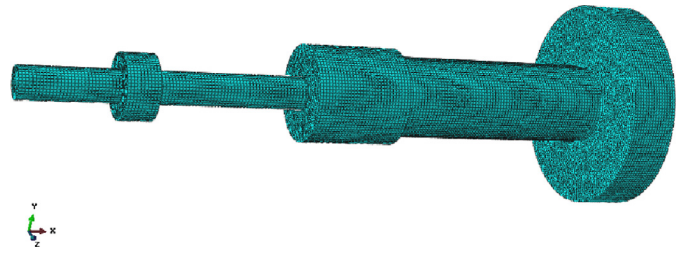


Fig. 19. Finite element model of the tri-axial impact test.

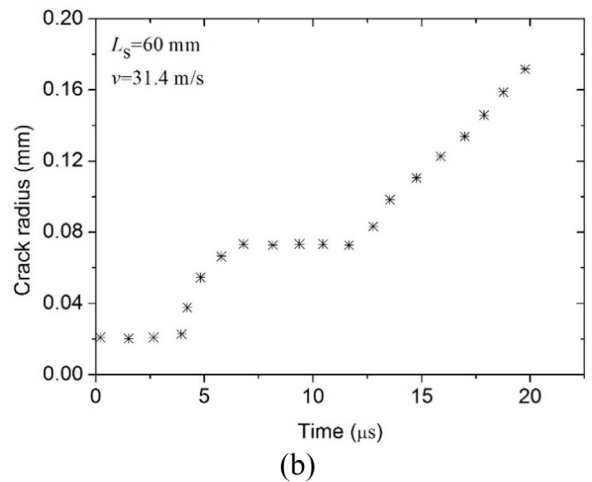
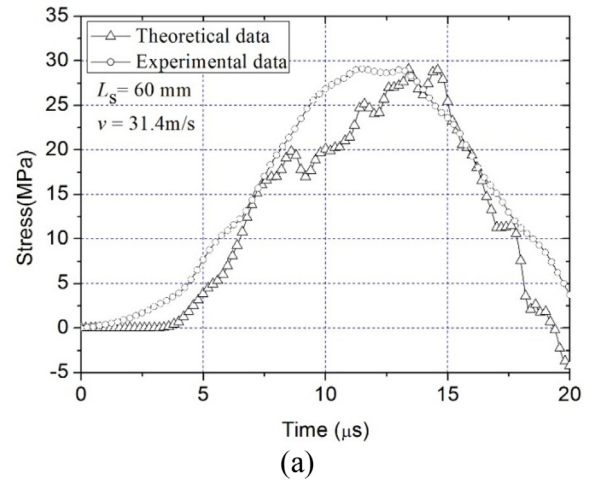


Fig. 20. (a) Comparison of the predicted and tested axial dynamic compressive stress-time curves, (b) Mean crack radius history of the PBX ($L_s = 60 \text{ mm}$ and $v = 31.4 \text{ m/s}$).

grows with time, giving reduced material moduli. The simulated results and the tested peak stress-velocity data in Table 2 are depicted in Fig. 21, showing a good agreement.

5. Conclusion

A series of impact tests was performed to obtain the influences of PBX ductility on its deformation and fracture behavior. Low-velocity impact tests were used to investigate the dynamic impact damage mechanisms and fracture modes. Various types of impact damages were observed at different impact velocities. The damage modes and fracture mechanisms were analyzed by SEM. For uniaxial impact loading, the main failure mode is trans-granular failure, while for multi-axial loading it changes to debonding and crystal cleavage. Microscopic examination reveals that the impact damage

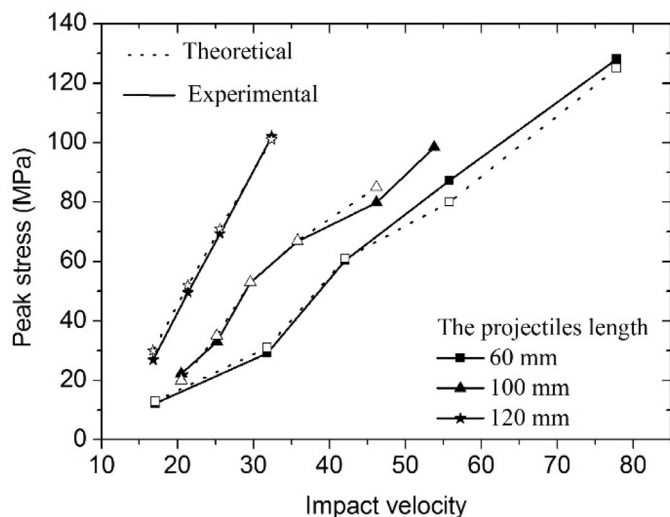


Fig. 21. Comparison between the simulated and the tested peak stress-impact velocity responses.

is not homogeneous. Impact loading induces many micro-cracks and micro-voids, which causes the density to decrease.

A statistical viscoelastic damage constitutive model of PBX was developed using the generalized energy release rate and dynamic fracture mechanics with viscoelastic effect. The constitutive model was implemented in ABAQUS to simulate the SHPB and the multi-axial impact experiments of the PBX. Some typical verification examples were presented to show the capacity of the model. The model can accurately predict the material behavior of low binder content PBX (e.g. PBX 9501, EDC37), and the damage behavior of high binder content PBX. The model provides a useful tool for identifying internal interacting mechanisms, which can not be distinguished using physical testing methods.

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